

LUNELIS

DATA-FLOW REPORT

Lunelis Data Flow

Where every photo, setting and byte comes from, goes and stays

PRODUCT	Lunelis 0.46.0
DATE	2026-10-09
STATUS	Draft for review (pilot of the documentation kit)
AUDIENCE	Maintainers, reviewers and anyone judging privacy
BASED ON	CLAUDE.md and src/lunelis (catalog, importers, raw, xmp, importing, edit, dupes, migrate, backups, jobs, recognize, geo, updater, gallery, darktable)

Statements in this document are read from the code and the changelog. Anything that could not be confirmed is listed under To check.

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CHAPTER 1

The big picture

Lunelis is one Python program on one PC. It indexes photos where they already sit, in folders called **sources** (the code calls them roots). It keeps its own records in a **data folder** and one SQLite **catalog**. This document follows the data: what is read, what is written, where it is kept, and the few places where anything leaves the PC.

44 schema migrations	9 background job kinds	4 internet hosts the app contacts	28 file types cataloged
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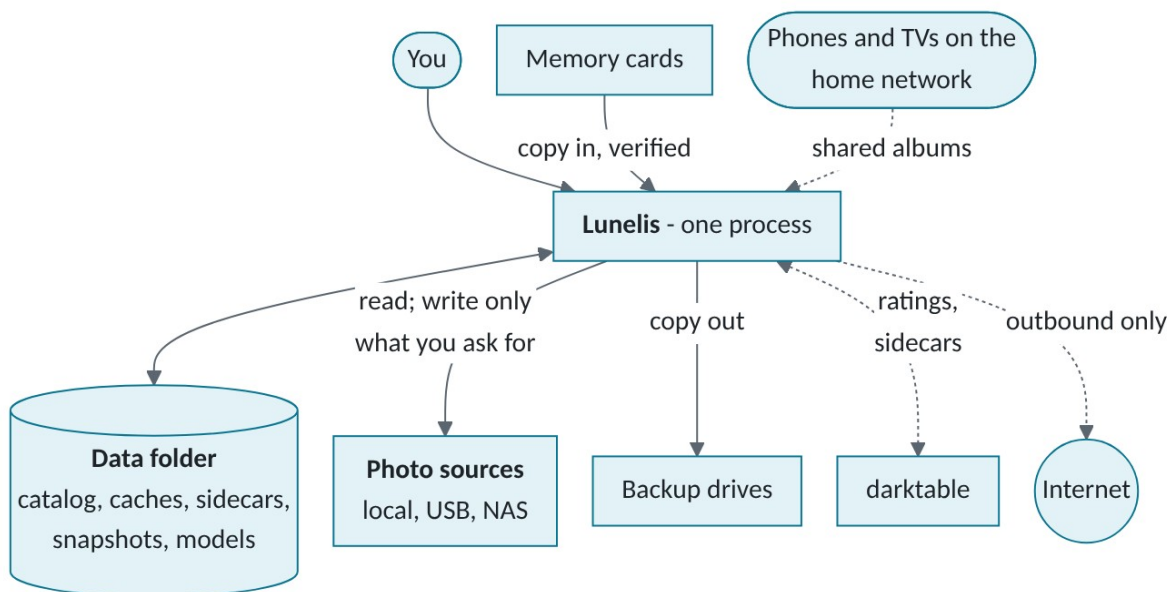


Figure 1. System context. Photos stay on their disks, Lunelis keeps its records in the data folder, and the internet is optional and outbound only.

- **Originals are read, not changed.** Lunelis writes into photo folders in a few cases only: an XMP sidecar next to a photo, files you import, files a migration copies, and renames into a **`_Lunelis Quarantine`** folder. Each case has its own chapter below.
- **Everything else lives in the data folder.** The catalog, thumbnails, edit previews, models, snapshots and the central sidecar store never go in a photo folder or the program folder (**`paths.py`**).
- **No photo data is sent anywhere.** The program contacts four hosts, all outbound. They see an internet address and a request, never a photo, a name or a location.

Each flow chapter that follows has the same parts: a diagram, a step table (step, from, to, data, stored in) and a box of rules that must not regress. The ends of the document hold the data inventory, the failure paths, what could not be confirmed (**To check**) and where the code and the project notes disagree (**Discrepancies**). Facts come from the code of version 0.46.0; file names are given so each one can be checked.

CHAPTER 2

Threads and trust boundaries

Lunelis is one operating-system process. Work is split across threads so the window never waits for a disk. A SQLite connection cannot cross threads, so every worker opens its own connection to the same catalog file (`open_catalog`: WAL mode, 20 second lock wait).

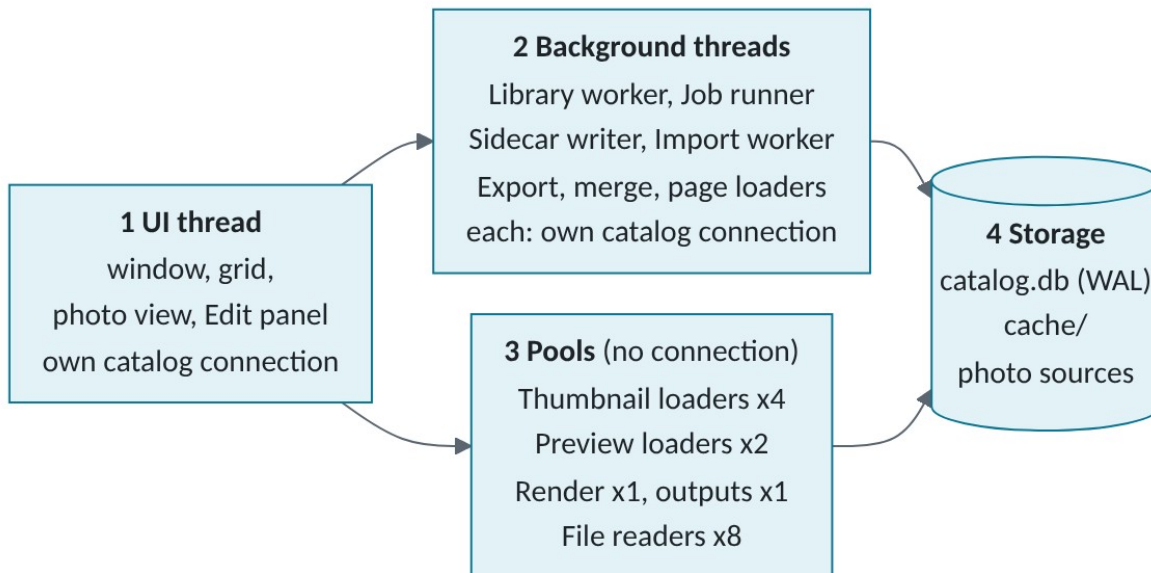


Figure 2. Four layers inside the one process. Layer 2 threads each hold a catalog connection; the pools only read files and cached images.

Worker	Runs as	What it does
Library worker	One <code>QThread</code> per pass	The library pass (below). Readers inside a step run 8 at a time
Job runner	One <code>QThread</code> per session	Takes queued jobs one at a time. 9 kinds in <code>jobs/engine.py</code>
Sidecar writer, import worker	A <code>QThread</code> per write or import	<code>export_pending</code> , 1.2 s after the last change; stage, verify and place card files
Thumbnail loaders	<code>QThreadPool</code> , 4 threads	Decode cached JPEGs for visible tiles. 1,500 pixmaps kept
Photo view, editor	Pools of 2, 1 and 1	Preview decode, live render, edit outputs in order
Family gallery	Server thread plus request threads	Only while an album is shared. One catalog connection per request

The **library pass** runs on one worker, in this order: scan, find moved files, read sidecars, read metadata, Takeout dates, burst stacks and timelapses, search index, find moved files by content, thumbnails, near-duplicate fingerprints, scene tags, faces, places, suggestions, damage check. A failed step is logged and the

rest still run. Every per-file worker calls `pace.breathe()`, which waits (at most 600 s) while a video plays, so playback is not starved.

Trust boundaries

Boundary	What crosses it	Guard in the code
You and the window	Keys, clicks, dialogs	Confirm before rating over 500 photos, downloads, emptying quarantine
App and photo sources	Reads; writes only as listed below	Read-only opens, temp name then replace, hash re-read, no rename or overwrite
App and data folder	Catalog, caches, models, snapshots	<code>paths.py</code> refuses a network path
App and memory cards	Reads, hashes; one delete you trigger	Clear the card re-hashes both copies
App and backup drives	Copies out, reads back	Volume serial, hashes, nothing deleted
App and other programs	XMP text, TSV files, darktable <code>luarc</code> , Takeout JSON	Text edits, newest change wins
App to internet	Update check, downloads, map tiles	HTTPS, SHA-256, opt-in, no photo data
Home network to app	Gallery requests	Home addresses, key per album, PIN, rate limits
Installer, updater	<code>setup.json</code> , release zip, swap script	<code>.sha256</code> , no <code>..</code> in the zip, old version restored
Model files	ONNX loaded into memory	SHA-256 at download only

What Lunelis writes into photo folders

In a photo folder	When	What happens to it
<code>name.EXT.xmp</code> sidecar	<code>beside</code> mode, or one exists and the update setting is on	Stays. Edited as text, never re-serialized
<code>.lunelis-*.xmp.tmp</code>	While a sidecar is written	Renamed over it, or deleted on failure
<code>.lunelis-importing-*</code> , <code>.lunelis-migrating-*</code>	While a file is copied in	Renamed to the real name after the flush
Imported and migrated files	An import or a migration	Stay. Names never change
<code>_Lunelis Quarantine/</code>	Copies and originals set aside (no Lunelis folder set)	Stays until you empty it

CHAPTER 3

Scanning folders into the catalog

The scan makes the **files** table match the disk. It reads directory listings only: names, sizes and times. It never opens a photo.

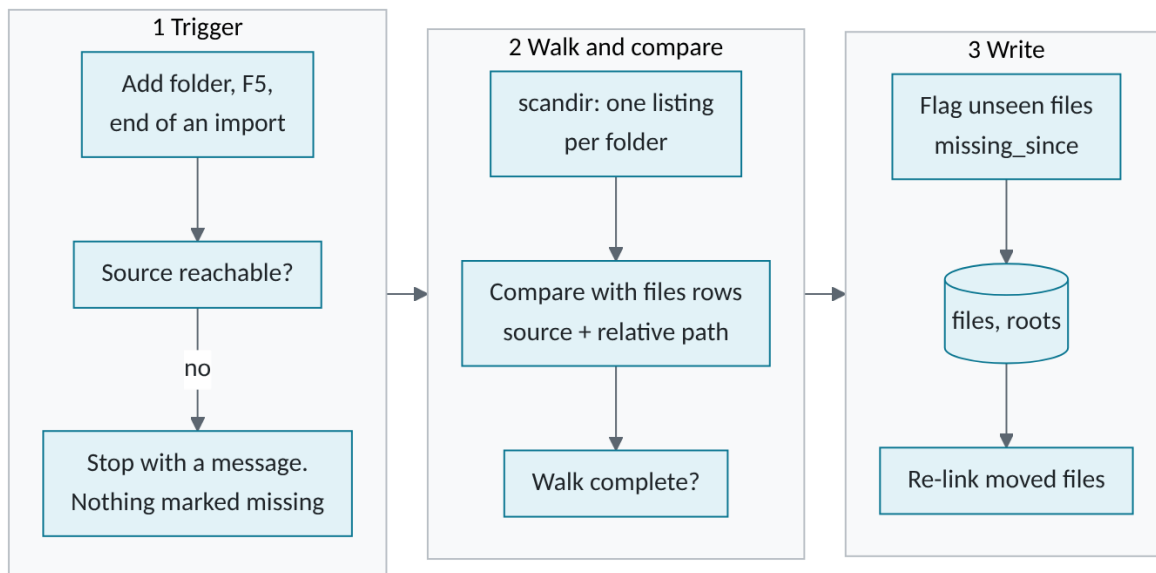


Figure 3. Scan. An offline source is refused rather than scanned as empty, and only a complete walk can flag files missing.

Step	From	To	Data	Stored in
1	You	Library worker	Source ids from Add folder, F5 or an import. One pass at a time	Memory
2	Library worker	Source	A folder-exists check. A missing source raises RootUnavailable , never "empty"	Nothing
3	Source	Library worker	Per folder: name, size, time, XMP names. Hidden, system and link entries skipped	Memory
4	Library worker	Catalog	New: insert. Size or time changed: update, clear hashes and thumbnail. Back again: clear the missing flag. Commit every 2,000 rows	files
5	Library worker	Catalog	Complete walk only: unseen files get missing_since (not quarantined, not in unreadable or skipped folders)	files, roots
6	Library worker	Catalog	A missing row takes over a new file with the same size and time. Rows with user data never merge	files, file_moves

WARNING • Scan rules that must not regress

- The scanner never deletes a **files** row. A missing file keeps its ratings, albums and faces.
- An unreachable source is never scanned as empty. A cancelled scan or an unreadable subfolder marks nothing missing under it.

CHAPTER 4

Reading metadata and sidecars

Sidecars are read first, then the files themselves. Readers run 8 files at a time and open everything read-only. Catalog writes stay on one thread, in batches.

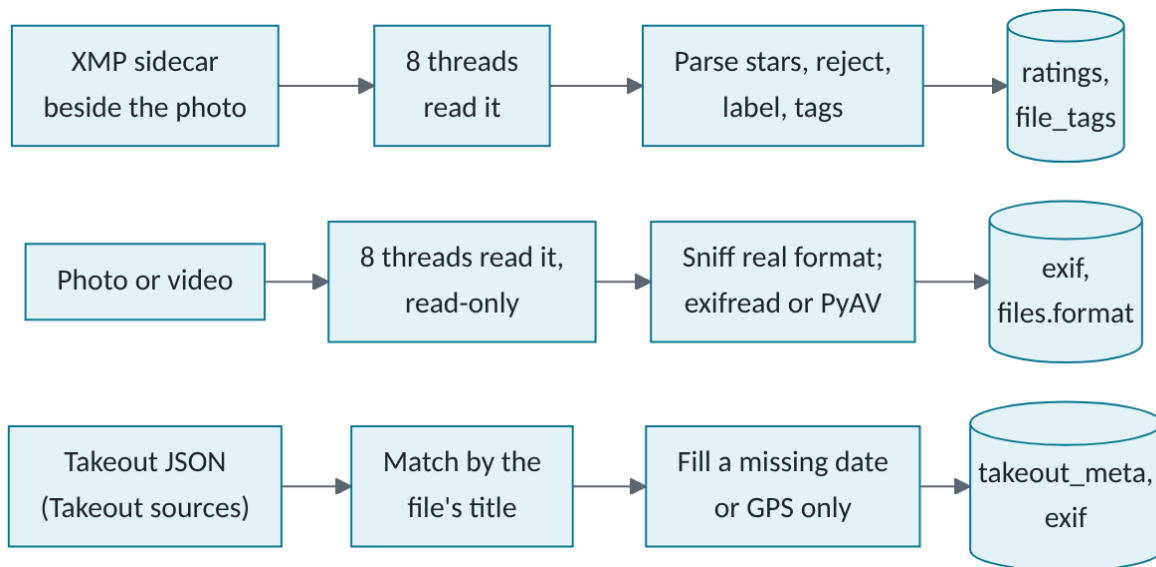


Figure 4. Three readers feed the catalog. Rows are committed in batches, so a stopped pass resumes where it left off.

Step	From	To	Data	Stored in
1	Sidecar	Library worker	Sidecars changed since the last sync, if no Lunelis change is waiting. Rating (-1 = reject), label, tags	Memory
2	Library worker	Catalog	A rating row only if one exists or values are not default. Tags only added. <code>lunelis:EditStack</code> parsed, never applied	<code>ratings, file_tags</code>
3	Original	Library worker	First bytes (real format), then EXIF. Maker notes only for real camera RAW. Videos through PyAV	Memory
4	Library worker	Catalog	Fields plus all EXIF as zlib JSON. Format and RAW flag set from the bytes. Batches of 200	<code>exif, files</code>
5	Library worker	Catalog	A failed read stores <code>read_error</code> , retried only when the file changes. An offline share leaves files pending	<code>exif</code>
6	Takeout JSON	Catalog	Date and GPS where the file has none. Never overrides EXIF	<code>takeout_meta, exif</code>

WARNING • Metadata rules that must not regress

- The extension is a guess. Decoders ask `files.format`, which comes from the bytes (`importers/formats.py`).
- No pyexiv2 (it cannot open non-ASCII paths on Windows). No maker notes on non-RAW files.

CHAPTER 5

Thumbnails and previews to the grid

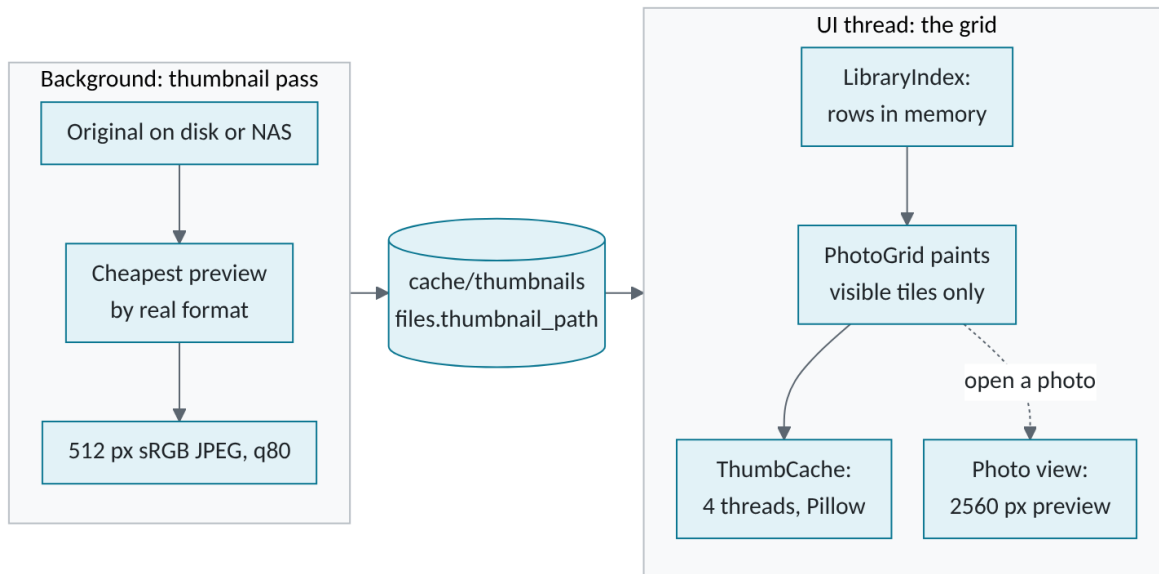


Figure 5. Thumbnails are made once, in the background. The grid only ever loads the small cached JPEGs.

Step	From	To	Data	Stored in
1	Original	Thumbnail pass	Smallest embedded JPEG of 512 px or more (RAW), camera JPEG preview, HEIC, or a video poster frame. Decoded in 8 threads	Memory
2	Thumbnail pass	Cache	Upright sRGB JPEG, 512 px long edge, quality 80, written to a temp name then replaced	cache/thumbnails/
3	Thumbnail pass	Catalog	Path or error text, plus a 64-bit fingerprint (dHash) taken from the picture in memory	files
4	Catalog	Grid	One query loads every row as plain tuples. Burst stacks collapse in Python	Memory only
5	Cache	Grid	A visible tile with no pixmap queues a load. Pillow decodes; the UI thread converts	Memory, 1,500 pixmaps
6	Original	Photo view	Biggest embedded preview, up to 2560 px. A full RAW decode only when you zoom past it	Memory

IMPORTANT • No RAW decode in the scroll path

The grid and the filmstrip load only cached JPEG thumbnails. Paint never touches the disk or the catalog. A RAW is decoded only on demand: in the editor, on an export, or when you zoom past the embedded preview. Deleting the thumbnail folder rebuilds it on the next pass; a file that fails keeps `thumb_error` until it changes.

CHAPTER 6

Ratings and tags to XMP sidecars

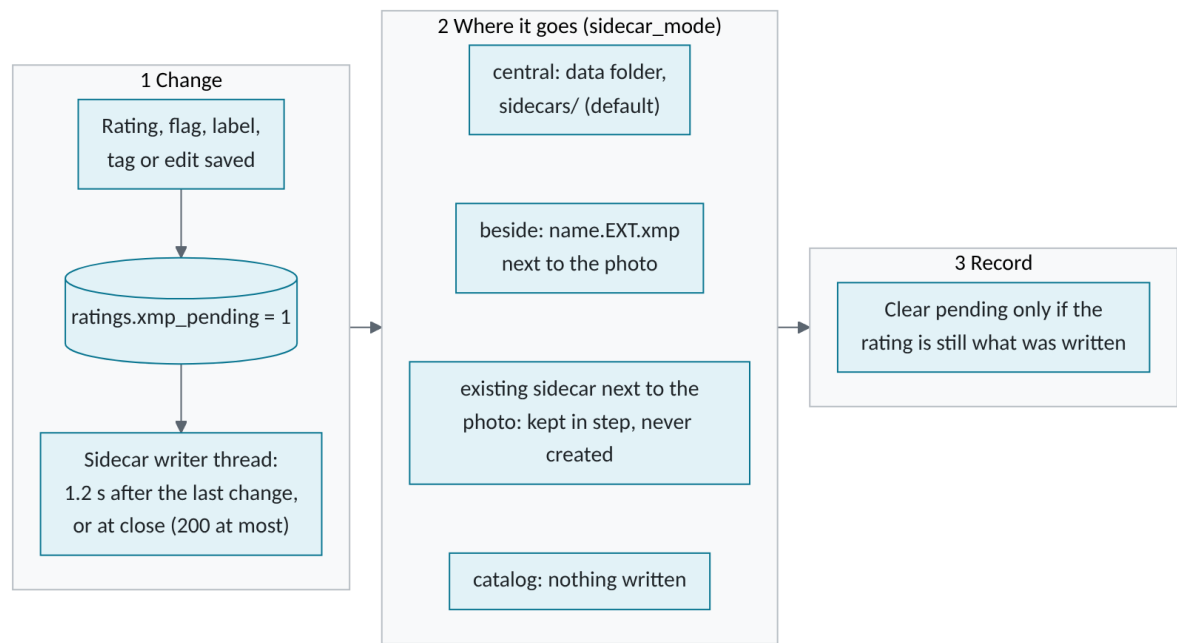


Figure 6. Sidecar export. One setting picks where the file goes, and an existing sidecar can be kept in step as well.

Step	From	To	Data	Stored in
1	You	Catalog	Stars 0 to 5, pick or reject, colour label, tag, or edit stack	ratings, file_tags, edits
2	Catalog	Sidecar writer	Every row with xmp_pending = 1: stars, flag, label, manual tags, edit text	Memory
3	Sidecar writer	Central store	sidecars/<id>-<source name>/<folder>/<file>.EXT.xmp	Data folder
4	Sidecar writer	Photo folder	Only in beside mode, or if a sidecar exists and the update setting is on (default)	Next to the photo
5	Sidecar writer	Sidecar	Text edit of changed fields only, checked well-formed, temp file then replace. A newer unwritten Lunelis change wins over the file	XMP text
6	Sidecar writer	Catalog	Pending cleared if stars, flag and label still match. A failure keeps it pending, with the error	ratings.xmp_error

WARNING · Sidecar rules that must not regress

- Never parse and re-serialize XMP: it renames darktable's namespace prefixes. Only changed values are edited as text.
- Reject is xmp:Rating = -1. A pick has no XMP field and never creates a sidecar. Adobe crs: settings are never written.

CHAPTER 7

Card import

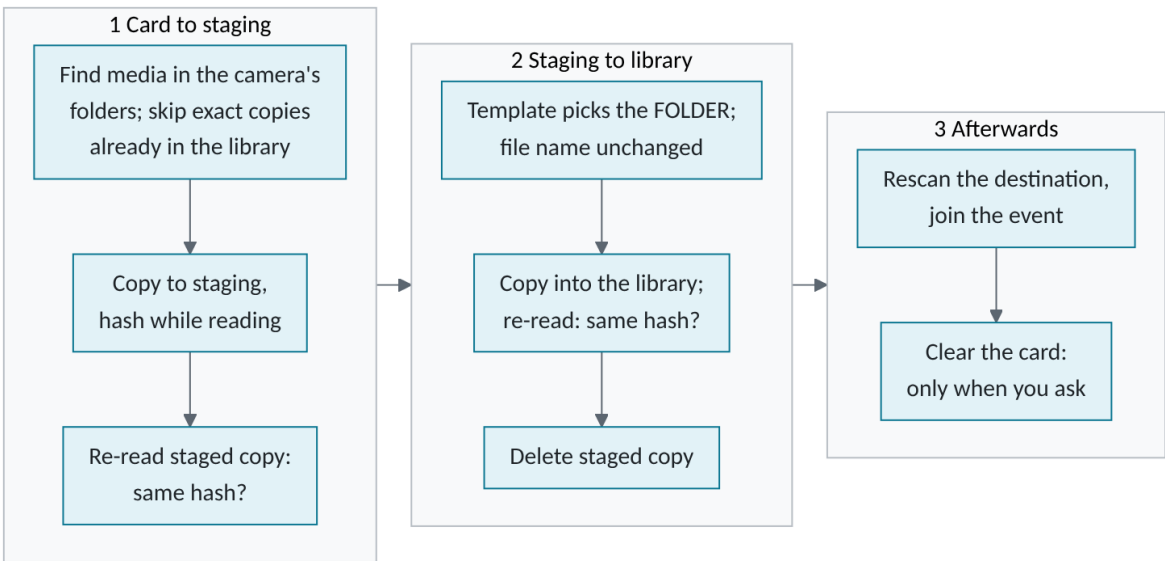


Figure 7. Card import. Two hash checks guard the copy, and the staged copy is deleted only after the library copy verifies.

Step	From	To	Data	Stored in
1	Card	Catalog	List of media, companions (Live Photo video) and sidecars, one row each	<code>imports, import_items</code>
2	Card	Staging	Copied to a <code>.lunelis-importing-*</code> temp name, hashed while read, flushed, renamed, re-read and compared	<code>staging/import-<id>/</code>
3	Staging	Library folder	Folder from the template and capture date. A different file with the same name goes to a sibling folder, ... (2)	Your destination
4	Library folder	Import worker	The new copy is re-read and compared with the card hash. Only then is the staged copy deleted	<code>import_items.state</code>
5	You	Card	Clear the card re-checks size, time and both hashes, then deletes card files. A changed file is left	The card

Staging is local while the disk keeps `import_local_reserve_gb` free (50 GB by default), then spills to the network staging folder if set, else the import waits. A pulled card or sleeping share sets `waiting`; the import resumes from `import_items`.

WARNING • Import rules that must not regress

- Imported files are never renamed. The template chooses folders only.
- "Already in the library" and "Clear the card" compare the **full SHA-256**; a sampled hash is never proof (audit LRA-001).

CHAPTER 8

Edits, previews and export

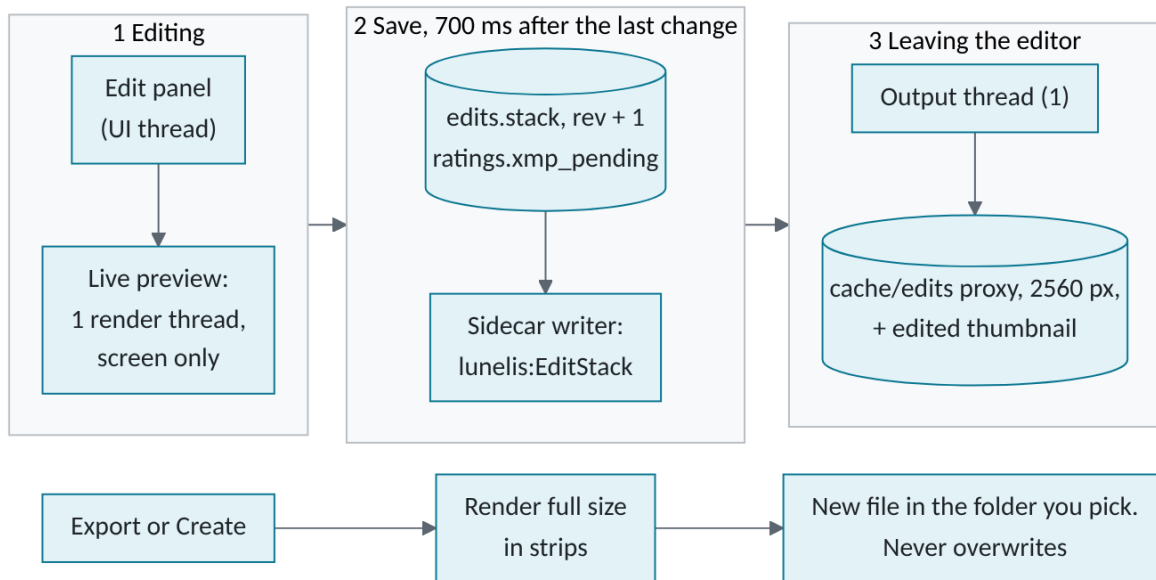


Figure 8. Edits are instructions, not pixels. Only the editor writes the cached look, through one ordered thread.

Step	From	To	Data	Stored in
1	Original	Edit panel	RAW through rawpy (half size when enough), JPEG at its own pixels. Float RGB, upright	Memory
2	Edit panel	Catalog	The stack as text: filter, sliders, crop, curves, masks. An identity stack deletes the row	edits, copies
3	Catalog	Sidecar	lunelis:EditStack , when sidecars are written (previous chapter)	XMP text
4	Edit panel	Cache	2560 px JPEG (quality 90) and a 512 px thumbnail with the edited look	cache/edits/, cache/thumbnails/
5	Original	Export	Full render in strips, sharpening, colour profile. EXIF rebuilt from the catalog: all, no location, or none	A new JPEG, TIFF or PNG
6	Edit panel	Cache	AI mask maps, made from the upright source and kept as PNG	cache/masks/

WARNING • Editing rules that must not regress

- The source file is never overwritten. An export is always a new file ("name (2).jpg" if the name is taken).
- Only the edit controller writes proxies and thumbnails. A late background write once put an edited look back on a reset photo.
- Exports use **apply_tiled** (strips). Never **apply** at full size: 60 MP in one piece needs several GB.

CHAPTER 9

Duplicates and quarantine

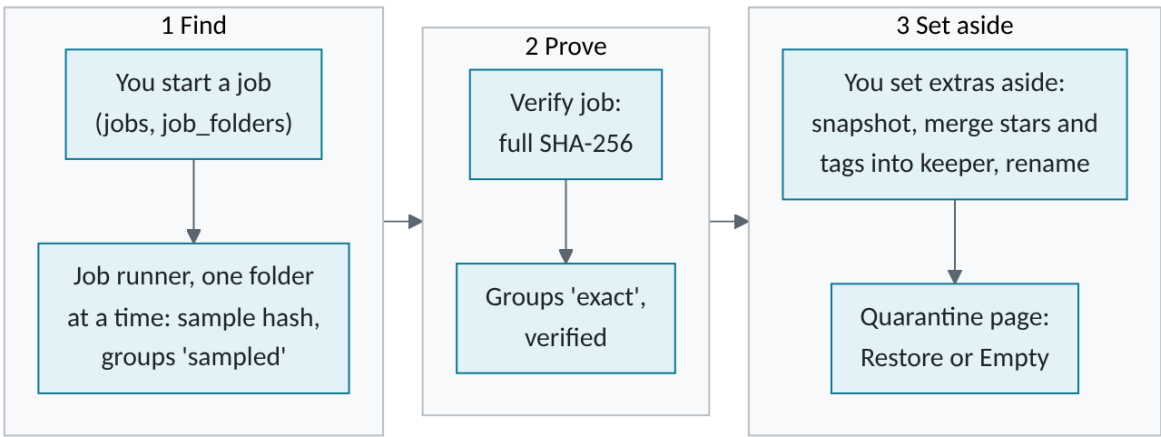


Figure 9. Duplicates. A sampled hash is a hint; only a full hash makes a group exact, and only exact groups can be set aside.

Step	From	To	Data	Stored in
1	You	Catalog	A scope, a schedule (now, idle, night), a speed limit. A folder is marked done only after its work is committed	jobs, job_folders
2	Original	Job runner	Sample hash: size plus 3 slices of 64 KB (about 200 KB a file), only for files sharing size and capture time	files.sample_hash
3	Original	Job runner	Verify job: every byte, SHA-256. Identical files form an exact group	content_hash
4	Catalog	Snapshot, catalog	Before a batch: a snapshot. Stars, tags, albums and sidecar of the extra go to the kept copy	backups/, ratings
5	Source	_Lunelis Quarantine	A rename on the same drive (copy, compare, remove across drives), sidecar included	files.quarantined_at
6	You	Recycle Bin	Empty: the kept copy must verify on disk. Local to the Recycle Bin; network removed after an extra tick	purged
7	Thumbnail pass	Catalog	Near-duplicates: a dHash per photo, grouped by complete linkage (4 bits, not RAW, same moment). Method similar	perceptual_hash

WARNING • Duplicate rules that must not regress

Only verified byte-identical groups can be set aside, never down to zero live copies. A job never hard-deletes; emptying quarantine is a user action with its own snapshot.

CHAPTER 10

Migration

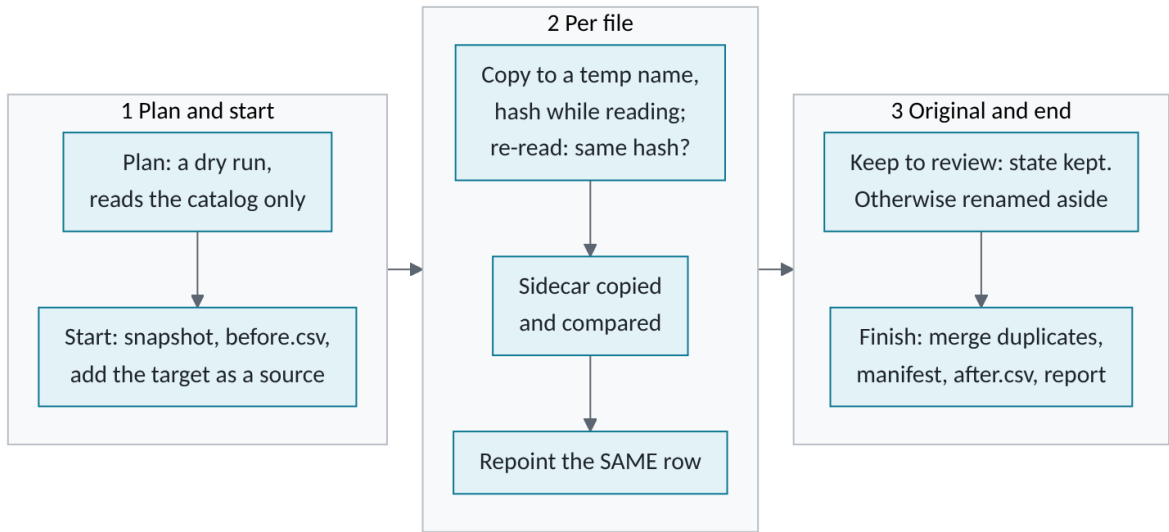


Figure 10. Migration. The same catalog row is repointed to the verified copy, so ratings, tags and thumbnails follow the file.

Step	From	To	Data	Stored in
1	Catalog	Plan	Per file: move, skip duplicate (an exact twin moves), or skip damaged (an intact copy exists). Folder from the template	migration_items
2	You	Logs folder	Start: snapshot, then before.csv listing every file in every source	Migration logs/
3	Source	Target	Copy to .lunelis-migrating-*, hashed while read, re-read and compared. A name clash goes to a sibling folder	Target folder
4	Source	Target	The sidecar is copied beside the photo and compared byte for byte	Target folder
5	Migration	Catalog	root_id and rel_path of the same row repointed, move logged, state copied	files, file_moves
6	Source	Set-aside	State done: renamed to the Lunelis folder's Trash, or _Lunelis Quarantine/migration-<id> on its own drive. kept in review mode, and the scanner skips it	Source drive
7	Catalog	Logs folder	End: manifest.csv (SHA-256 of each copy), after.csv, an accounted-for report	Migration logs/

- WARNING · Migration rules that must not regress**
 - A plan changes nothing on disk. Never delete, overwrite or rename. Originals are released only when the report finds nothing missing.
 - A crash is safe: before step 5 the source is untouched; after it a resume only finishes the original.

CHAPTER 11

Backups and catalog snapshots

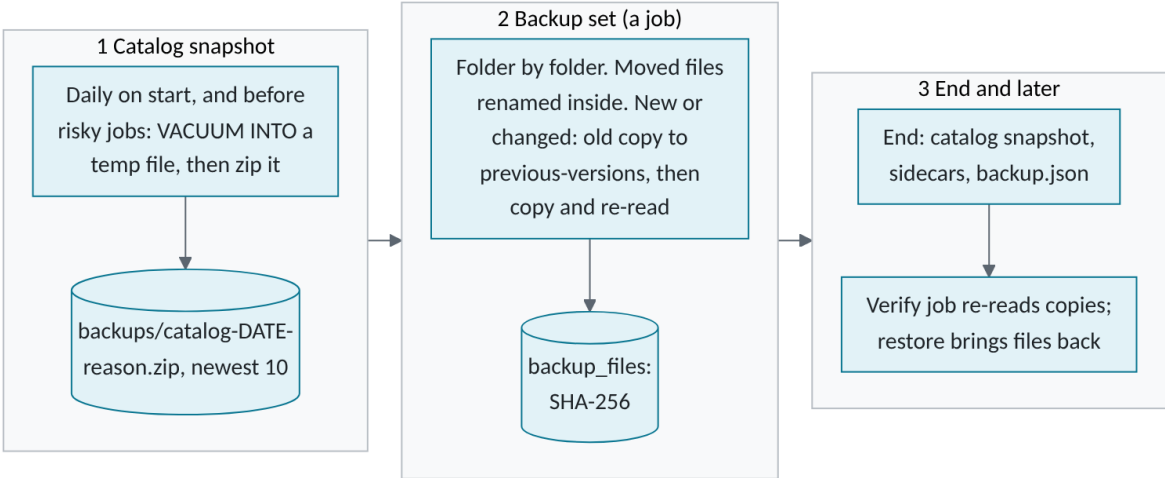


Figure 11. Two separate safety nets. Snapshots protect the catalog; backup sets protect photo files.

Step	From	To	Data	Stored in
1	Catalog	Snapshot folder	A consistent copy made while the catalog stays open, zipped. Newest <code>catalog_backups_keep</code> (10) kept	<code>backups/</code>
2	Library	Backup drive	New or changed files, hashed while read, re-read, compared. A mirror of each source's folders	<code><backup>/<id>-<name>/</code>
3	Backup drive	Backup drive	A changed file's old copy moves to <code>previous-versions/<time>/</code> inside the backup. Nothing is deleted from a backup	Backup drive
4	Job runner	Catalog	Size, time, SHA-256 per copy. The first backup fills <code>files.content_hash</code> . At the end: a snapshot (5 kept), sidecars, <code>backup.json</code>	<code>backup_files, _Lunelis/</code>

Step	From	To	Data	Stored in
5	Backup drive	Library	Restore: missing or damaged files come back (a damaged one is quarantined first). The catalog returns at the next start	restore-pending.json
6	Job runner	Catalog	Verify job re-reads each copy against its hash. A missing drive makes the job wait; a full one pauses it	backup_files

WARNING · Snapshots and restores

A snapshot is taken before every job that moves or removes files, and before every schema upgrade. All snapshots share the newest 10, so an old one is deleted as newer ones arrive. Picks, events, albums, faces and edits live only in the catalog, so snapshots protect them.

CHAPTER 12

The darktable bridge

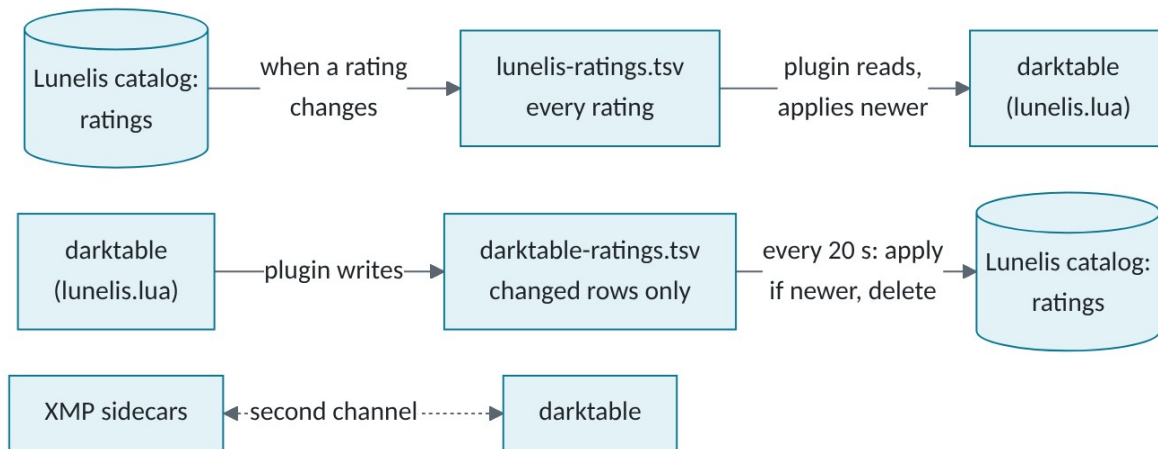


Figure 12. darktable bridge. Two TSV files in one exchange folder; the newest change wins. XMP sidecars are a second channel.

Step	From	To	Data	Stored in
1	You	darktable	Install copies <code>lunelis.lua</code> into darktable's <code>lua</code> folder and adds one <code>require</code> line to <code>luarc</code>	<code>%LOCALAPPDATA%\darktable\</code>
2	Catalog	Exchange file	Every rating: full path, stars or -1 for reject, label, UTC change time	<code>lunelis-ratings.tsv</code>
3	Exchange file	darktable	At darktable start-up, on a view switch and on "Sync with Lunelis": only rows newer than its <code>last_applied</code> preference	darktable's library
4	darktable	Exchange file	Same moments, and when darktable closes: rows changed since its last snapshot	<code>darktable-ratings.tsv</code> , <code>darktable-snapshot.tsv</code>
5	Exchange file	Catalog	Matched by full path, case-insensitive. A row older than the catalog's change is skipped	<code>ratings</code>
6	Catalog	Sidecar	An applied change queues a sidecar write like any rating	See the sidecar chapter

The exchange folder is `darktable/` in the data folder unless you set another. The Lunelis side checks every 20 seconds on the UI thread, and is cheap when nothing changed.

WARNING · darktable rules that must not regress

- The plugin edits no photo, no darktable history and no XMP file directly. darktable writes its own sidecars as usual.
- Newest change wins. Picks never cross, because darktable has none. darktable may hold several colour labels; Lunelis keeps one and never removes the extras.

CHAPTER 13

Models and recognition

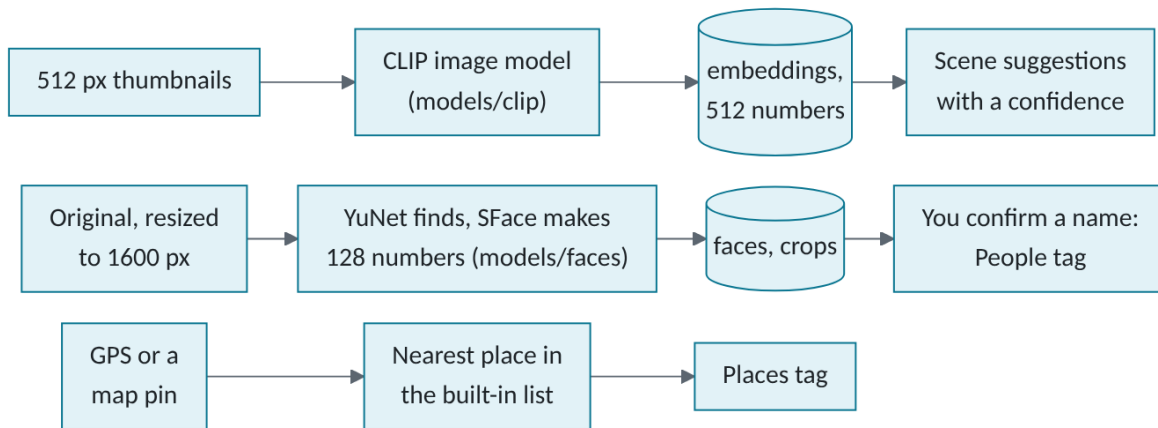


Figure 13. Recognition. Everything runs on the CPU in this process. Each path reads different pixels, and only names you confirm become tags.

Step	From	To	Data	Stored in
1	Internet	Models folder	You confirm. Each file is checked against its SHA-256 and size, then moved into place	models/
2	Cache	CLIP	Thumbnails only: never a RAW or the NAS. Scored against 40 labels: 25 % or more, 3 per photo at most	embeddings, file_tags
3	Original	YuNet, SFace	Upright copy, 1600 px at most. A face needs score 0.8 and 2.5 % of the long edge. Videos skipped	Memory
4	YuNet, SFace	Catalog	Box as fractions, 128-number fingerprint, model name, 160 px crop. Each photo once per model	faces, cache/faces/
5	Catalog	Catalog	Suggested to a named person at cosine 0.40, grouped at 0.45. Only a face you confirm tags the photo	faces, file_tags
6	Catalog	Catalog	Nearest of 31,735 listed towns: named within 40 km, country only within 250 km. A pin beats GPS	photo_places, locations
7	Original	Mask maker	Subject and sky masks from the upright source, cached as PNG. Ask matches your words to stored fingerprints	cache/masks/

WARNING • Recognition rules that must not regress

Everything that understands pixels goes through the **Recognizer** interface, so it can move to an external service later. Suggestions carry a confidence, and every normal tag query filters **confidence IS NULL**.

CHAPTER 14

Network use

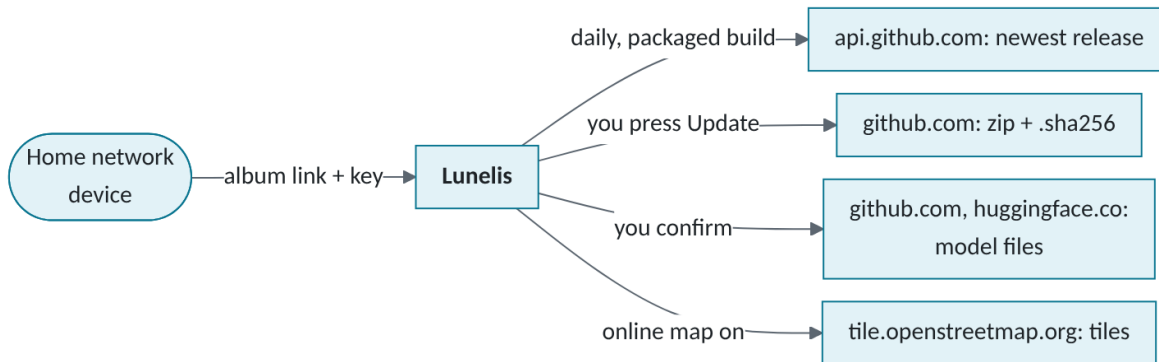


Figure 14. Network use. Four outbound hosts, each started by a setting or a click, and one inbound path that stays on the home network.

Destination	Why	When	What is sent	Default
api.github.com	Is a newer release out?	8 s after start (packaged build, last check over 20 h ago), or the Check button. Setting <code>update_check</code>	GET; internet address, <code>Lunelis-updater</code>	On
github.com (releases)	Update zip and <code>.sha256</code>	You choose Install	GET; internet address	Off
github.com (OpenCV, rembg)	Face and subject-mask models	You confirm a download	GET; <code>Lunelis</code>	Off
huggingface.co	CLIP and sky-mask models	You confirm a download	GET; <code>Lunelis</code>	Off
tile.openstreetmap.org	Map pictures	Map shows tiles (<code>map_online</code>); each tile fetched once	Zoom, column, row; <code>Lunelis/x.y.z</code>	Off
www.openstreetmap.org	Show a place in your browser	You click, and "location opens" is the browser	Coordinates, in a URL	Off
Home network, default port 8735 (in)	Family gallery	While an album is shared	1600 px JPEGs without EXIF	Off

A search of `src/lunelis` for `urllib`, `QNetworkAccessManager`, `socket` and `http` finds only the files above. There is no account, no telemetry and no crash upload; Help > Report a problem shows text for you to copy. Every model download and the update check send the program name in the `User-Agent` header and nothing else you could identify; the host sees your internet address.

NOTE · What comes down is checked, but only so far

- Model downloads are hashed on arrival and deleted on a mismatch. Existing model files are later accepted by size only.
- An update needs a matching `.sha256`. It comes from the same release as the zip, so it shows the download is intact, not who made it. The build is unsigned.
- Map tiles are decoded as images before saving; they are not hashed.

WARNING · Family gallery

Plain HTTP on all addresses of the PC (`0.0.0.0`). A request from outside the home ranges is refused first. Then: a random key per album, an optional PIN (PBKDF2, 200,000 rounds), 5 wrong PINs lock an address for 10 minutes, 20 lock an album for an hour, at most 32 connections, 1600 px copies with no EXIF. Home-network traffic is not encrypted.

The model downloads in full. All four sets come to about 414 MB. Each file is pinned by SHA-256 and size in the code (`recognize/clip.py`, `recognize/faces.py`, `edit/ai.py`).

Model	Files and sizes in bytes	Source	Saved in
YuNet and SFace (faces)	Detector 232,589; recognizer 38,696,353	github.com, OpenCV model zoo at a pinned commit	<code>models/faces/</code>
CLIP ViT-B/32 (scenes, Ask)	Image model 89,117,001; text model 64,504,507; two word lists 862,328 and 524,619	huggingface.co, an ONNX copy of OpenAI CLIP	<code>models/clip/</code>
Silueta (subject mask)	One file, 44,173,029	github.com, rembg release v0.0.0	<code>models/</code>
Skyseg (sky mask)	One file, 175,997,079	huggingface.co, a published skyseg model	<code>models/</code>

CHAPTER 15

Data inventory

Data	Where it lives	Kept for	Leaves the PC?
Photos and videos	Your source folders	Until you remove them	Only by your action: export, import, backup, shared album
Catalog: ratings, picks, tags, albums, events, faces, settings, EXIF	<code>catalog.db</code> in the data folder	For ever	No. In snapshots and backup sets
Edit stacks, virtual copies	<code>edits</code> , <code>copies</code> ; <code>lunelis:EditStack</code> in sidecars	For ever	Sidecar text goes into backups
Thumbnails, edit previews, masks	<code>cache/</code>	Until cleared	No
Sidecars	<code>sidecars/</code> (central) or beside the photo	For ever	In backup sets
Face fingerprints, names, crops	<code>faces</code> , <code>people</code> , <code>cache/faces</code>	For ever	No. In snapshots, so a shared snapshot shares them
Scene fingerprints, suggestions	<code>embeddings</code> , <code>file_tags</code>	For ever	No
Place tags, pins	<code>photo_places</code> , <code>locations</code>	For ever	Tags in sidecar text; pins no
Models	<code>models/</code>	Until removed	They come in only
Catalog snapshots	<code>backups/*.zip</code> or your folder	Newest 10 of all kinds	In backup sets (5 kept)
Set-aside files	<code>_Lunelis Quarantine</code> , or Lunelis folder Duplicates and Trash	Until you empty them	No
Staging, update files	<code>staging/</code> , <code>updates/</code>	Until placed or cleaned	The update zip comes in
Map tiles	<code>map_tiles/</code>	For ever	They come in only
Gallery copies	<code>gallery_cache/</code>	Until cleared	Served on the home network
Log	<code>logs/lunelis.log</code> , 5 files of 2 MB	Rotated	No. May name folders and files
Location and set-up records	<code>location.json</code> , <code>setup.json</code> , Windows Run and Uninstall keys	Until changed	No

CHAPTER 16

Failure paths

What fails	What the program does	What the user sees
NAS or drive offline	The scan refuses the source. Metadata and thumbnail passes leave its files pending. Jobs go to waiting and retry every 60 s. An import waits	"Folder unavailable"; photos marked OFFLINE (checked every 60 s); "Waiting for ... to come back online"
Staging disk full	The import pauses; no file is half placed	"Out of staging space"
Backup drive full or removed	The job pauses or waits. Nothing is deleted from the backup	The set's status line says why
Power cut or crash	Scans keep committed batches. A job resumes at its first pending folder; an import from import_items ; a migration at its last safe state	"Resuming after an interruption"
Catalog damaged or empty	Start-up will not open it, and never starts silently empty	A box: restore the newest snapshot (bad catalog kept beside it), start empty, or quit
Catalog from a newer version	Refused, untouched	A message to install the newest Lunelis
location.json unreadable	The default data folder is used for now; nothing moves	A warning naming the file
A sidecar cannot be written	The change stays pending in the catalog and is retried (60 s; 5 min after an error)	"Couldn't write N sidecar(s) - kept in the catalog"
A photo cannot be decoded	thumb_error or read_error is stored; skipped until it changes	A placeholder tile; a row in Damaged files
A model is missing	The step is skipped silently; Ask falls back to word search	A "Download and turn on" button in Settings
A download fails its hash	The partial file is deleted; nothing is installed	"Didn't match its checksum - nothing was installed"
The update cannot be swapped in	The script puts the old folder back and starts it	A warning on the next start
A card is pulled mid-copy	The import waits; it resumes when the same card (volume serial) is back. Staged files are kept	"Waiting for the card to be inserted again"
A migration copy does not match	The copy is deleted and the file fails; the original is untouched	A failed item in the Migrate page
Map tiles cannot be reached	Tiles already saved still draw. The dots always work	A one-time note on the Map page
The gallery port is busy	The server does not start	"The family gallery couldn't start on port N"
One library-pass step errors	It is logged; the other steps still run	"Finished, but these stopped on an error: ..."

REFERENCE

To check

- **Start-up scan.** No timer or start-up call that starts a scan was found in `ui/main_window.py`. Scans start from Add folder, F5, Library status, Settings, an import and the migration wizard. Confirm this is intended.
- **Redirect hosts.** The code names `github.com` and `huggingface.co`. The final download servers after their redirects are not in the code.
- **Model files at rest.** `available()` in `recognize/clip.py` and `recognize/faces.py` accepts a file of the right size without re-hashing it.
- **Update trust.** The `.sha256` and the zip come from the same release. Whether signing is planned is not stated in the code.
- **Gallery exposure.** The server binds to all addresses and checks the client address. How Windows Firewall treats it was not examined. PIN and cookie travel unencrypted.
- **Pending flag race.** After a sidecar write, `xmp_pending` is cleared when stars, flag and label still match (`xmp/sync.py`). A tag or edit saved during that write is not compared. Not tested.
- **RAW decode in the photo view.** Preloading a neighbour that is an edited RAW with no cached proxy decodes it at half size (`ui/detail_view.py`). That is outside the grid and filmstrip, but it is a RAW decode while paging.
- **Proxy settings.** Whether `QNetworkAccessManager` follows the Windows proxy settings was not examined.
- **darktable plugin.** `CLAUDE.md` says the Lua script was tested only under a Lua 5.4 stand-in, never in a real darktable. Not re-checked here.
- **Outside the program.** The Inno Setup installer and the CI workflows are not in `src/lunelis` and are not covered.
- **Full data disk.** No code was found that handles a full data-folder drive in particular. A failed write inside a library-pass step is logged and the other steps run; other pages were not traced.
- **Request headers.** Qt may add default headers (such as `Accept`) to a tile request beyond the `User-Agent` set in `ui/map_view.py`. Not examined.
- **Paths in logs.** Status messages and errors go to `logs/lunelis.log` and can name folders and files. The problem report copies the last 200 log lines. Whether to scrub paths is not decided in the code.
- **Daily snapshot.** The "daily" snapshot is checked once, 3 s after start-up (`_daily_backup`). No timer was found, so a tray program left running for days snapshots again only at its next start, or before a risky job.
- **Speeds.** Rates quoted in `CLAUDE.md` were measured on its author's library and not re-measured.

REFERENCE

Discrepancies

Where	The documents say	The code does
<code>docs/_tools/architecture.mmd</code>	"SQLite schema v37". The jobs engine runs "dust maps". One box holds "Import + Autopilot, migrate, photo backups"	<code>MIGRATIONS</code> ends at version 44. There is no dust job kind (<code>dust_view.py</code> uses a page loader). Import and autopilot have their own threads; migrate and backups are job kinds
<code>CLAUDE.md</code> , Architecture	Edits are "an instruction stack in XMP"	The stack is written to the sidebar but never read back: <code>import_sidecars</code> ignores <code>lunelis:EditStack</code> , and nothing reads the central store. Only snapshots and backups restore edits
<code>CLAUDE.md</code> , thumbnail and metadata rules	Video thumbnails "not yet"; CR3 and video have no EXIF reader. Cache path <code><id//1000>/<id>.jpg</code>	Videos get a poster-frame thumbnail (PyAV) and a metadata probe; only CR3 has no reader. <code>cache_rel_path</code> zero-pads: <code>0012/12819.jpg</code>
<code>CLAUDE.md</code> , non-negotiables	"Lunelis's own files never go in the photo library"	Each source can hold <code>_Lunelis Quarantine</code> , and <code>.lunelis-*</code> temp files while writing. A Lunelis folder you choose can hold exports, trash and logs
<code>CLAUDE.md</code> , backup rule	A snapshot is taken before every job that moves or removes files	True, but Duplicates, Near-duplicates and Quarantine snapshot into <code>data/backups</code> , not the Settings folder that daily snapshots use. Start-up recovery also looks only in <code>data/backups</code>
<code>catalog/schema.py</code> , upgrade snapshot	The pre-upgrade snapshot "prune[s] nothing" and is "the way back"	It prunes nothing when made, but every later snapshot keeps only the newest <code>catalog_backups_keep</code> (10), so it is deleted after about ten more

Where	The documents say	The code does
CLAUDE.md scanner rules; geo/places.py	A change nulls three columns. About 34,000 places	The update also clears sample_hash and thumb_error. The list holds 31,735 rows
CLAUDE.md, Quarantine rules	empty() is the only hard removal	Clear the card (importing/ingest.py) also deletes, from the card, after a double hash check and your confirmation